

ServiceNow IT Equipment Request System

Project Design & Implementation Guide

1. Project Overview

Company Background: Northstar Technologies is a fictional professional services company with approximately 250 full-time employees distributed across three offices throughout Florida. The organization has experienced rapid growth over the past two years, significantly increasing the demand for standardized internal IT processes.

Business Problem: The organization currently manages IT equipment requests through email and spreadsheets, resulting in duplicate requests, inconsistent approval processes, delayed fulfillment, limited request tracking, and poor visibility throughout the fulfillment lifecycle.

Proposed Solution: This project implements an IT Equipment Request System within ServiceNow that centralizes equipment requests through the Service Catalog while automating approvals, fulfillment activities, request tracking, and lifecycle management.

Expected Outcome: The completed solution provides employees with a centralized request process while improving operational efficiency and visibility for managers and IT staff.

2. Users & Roles

The IT Equipment Request System is designed around four primary user personas, each with distinct responsibilities throughout the equipment request lifecycle.

Persona	Description	Primary Responsibilities
Employee	Company employee requesting standard IT equipment.	Submit requests, monitor request status, provide additional information if needed.
Manager	Supervisor responsible for reviewing and approving employee equipment requests.	Review requests requiring approval, approve or reject requests, monitor pending approvals.
IT Fulfillment Agent	IT staff responsible for fulfilling approved equipment requests.	Receive fulfillment tasks, prepare the equipment, update request progress, complete assigned tasks.
System Administrator	System administrator that is responsible for maintaining the equipment request application.	Configure the application, manage users and roles, troubleshoot issues, maintain workflows and system configuration.

3. End-to-End Workflow

The IT Equipment Request System follows a structured workflow that guides an equipment request from initial submission through fulfillment and completion. The following phases describe the expected lifecycle of a standard equipment request.

Request Submission

Employees initiate equipment requests through the Service Catalog by selecting one of two available catalog items: **Standard Workstation Package** or **Additional Equipment Request**. The Standard Workstation Package provides employees with a pre-approved Windows or macOS workstation that includes the essential equipment required to begin work and proceeds directly to fulfillment without managerial approval. Employees requiring additional equipment beyond the standard package may submit an Additional Equipment Request, where they can select one or more approved accessories, provide a business justification, specify a requested completion date and priority, and include any additional comments relevant to the request.

Request Validation

After a request is initiated, the system verifies that the employee is eligible to submit the selected request and that the request satisfies the organization's equipment request policies before it can be submitted. Employees may submit only one active Standard Workstation Package request and cannot request another package while one is already assigned or awaiting fulfillment. Access to the Additional Equipment Request requires the employee to have a fulfilled workstation package. Employees who are not yet eligible may view the catalog item but are shown an explanatory message and prevented from submitting the request.

For Additional Equipment Requests, the system displays the employee's currently assigned equipment and any pending equipment requests as read-only summaries. Only equipment that is neither assigned nor currently pending remains available for selection. The employee must select at least one eligible equipment item, provide a business justification, and specify a valid Needed By date, which the system uses to automatically determine the request priority. Requests due within three calendar days are assigned an Urgent priority, while later requests receive a Standard priority. Urgent requests require the employee to provide a reason for the requested completion date, and dates in the past are not permitted.

Approval Evaluation

After validation, the system determines the request's approval path based on the selected catalog item. Standard Workstation Package requests are pre-approved and proceed directly to fulfillment, while Additional Equipment Requests are routed to the employee's manager. Employees without an assigned manager cannot submit an Additional Equipment Request until their user information has been corrected.

Manager Approval

The employee's manager receives an approval request containing the submitted equipment selections and request details. The manager may review the information but cannot modify the employee's submission. The Additional Equipment Request must be approved or rejected as a whole, and approval comments are optional. If the manager rejects the request, an explanation is required before the request can be closed, and the employee is notified of the decision.

Request Fulfillment

Approved requests proceed to fulfillment, where a series of Catalog Tasks is automatically created and assigned to the IT fulfillment team. These tasks include verifying equipment availability, preparing the approved equipment, and coordinating its delivery to the employee. Once all Catalog Tasks have been completed, the fulfillment process is considered complete and the request proceeds to the completion phase.

Request Completion

After all fulfillment activities have been completed, the system marks the request as complete, updates the employee's assigned equipment records to reflect the newly issued equipment, and notifies the employee that the approved equipment has been issued.

4. Data Model

Overview

The application uses standard ServiceNow tables to manage users, requests, approvals, and fulfillment activities. These tables support the catalog request lifecycle without requiring unnecessary customization. A custom Equipment Assignment table is introduced to maintain a persistent record of equipment issued to employees after fulfillment. Unlike request and fulfillment records, which document the approval and delivery process, this table represents the employee's current and historical equipment assignments and provides a reliable source for reporting and equipment management.

Tables

Table	Type	Purpose
User	Standard	Stores employee and manager information used during request submission and approval routing.
Request	Standard	Represents the overall catalog request submitted by the employee and tracks the request throughout its lifecycle.
Requested Item	Standard	Represents the employee's equipment request and tracks its approval and fulfillment status.
Approval	Standard	Records manager approval decisions associated with requested items and tracks the approval outcome.
Catalog Task	Standard	Represents the fulfillment work assigned to the IT fulfillment team and tracks task completion.
Equipment Assignment	Custom	Maintains employee equipment assignment records after request fulfillment has been completed.

Equipment Assignment Table

Field	Type	Purpose
Employee	Reference	Identifies the employee receiving the equipment.
Equipment Type	Choice	Identifies the category of equipment being assigned to the employee.
Status	Choice	Indicates whether the equipment is currently Assigned or Returned .
Assigned Date	Date	Records the date on which the equipment was issued.
Returned Date	Date	Records the date on which the equipment was returned.
Requested Item	Reference	Identifies the request that resulted in the equipment assignment.

5. REST Integration

Upon successful fulfillment of an equipment request, the application performs an outbound REST integration with an external Equipment Reporting Service. After the employee's equipment assignment has been updated, the flow sends an HTTP POST request containing the employee's ID, name, equipment type, assignment date, and originating request number. The external service, implemented as a lightweight Node.js and Express application deployed on Render, receives the request and temporarily stores it in memory for demonstration purposes before returning a JSON confirmation response. ServiceNow evaluates the HTTP status code and response body to verify that the integration was successful. The reporting service also exposes a GET endpoint that returns all received equipment assignments and supports filtering by employee IDs through query parameters, allowing assignment data to be retrieved and verified during the project demonstration.

6. Project Walkthrough